

Understanding the Contributions of the Milky Way and M31 to the Merger Remnant Shape (Research Assignment 6)

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1. INTRODUCTION

The area of galaxy evolution this research will investigate is the impact that galaxy mergers have on dark matter halos. This research will specifically look at mergers in the **Local Group** which is the gravitationally bound systems of galaxies of which the Milky Way is a member. Dark matter is believed to be dispersed throughout the universe in an almost web-like structure with a vast system of filaments. **Dark matter halos** are the areas in the galaxy where there is a high density of dark matter. The overall shape that these halos take is the **Halo Shape**, and the rotation of the halo is the **Halo Spin**. The Halo Shape is **Triaxial**, which means it is described by its three axes. If one of the axes is proportionally smaller or larger, the shape of the halo can be **Prolate**, or **Oblate** ellipsoids. These halos are theorized to come about when multiple filaments intersect with one another. These dark matter halos are significant because they are where galaxies are able to form [C. Frenk & S. White \(2012\)](#).

A **Galaxy** is a system of stars that are gravitationally bound and have physical properties that cannot be explained by just the stars alone, we define galaxies as being made of stars and dark matter. Because these dark matter halos are thought of as the basic unit of cosmological structure, understanding dark matter halos is intrinsic to understanding cosmology. Researching the structure of dark matter halos, as well as halo formation and halo growth can give key insights into **Galaxy Evolution**, which is the formation and life cycle of a galaxy. Understanding halo growth requires understanding how these structures are impacted when galaxies merge.

The main understanding of dark matter halo mergers comes from cosmological simulations of mergers in near perfect settings. So far, the general understanding finds that the size and spin of the halo remnant are dependent on the energy of the orbit and the angular momentum of the orbit respectively. The shape of the remnant is primarily dependent on the initial separation between the two halos as well as the spin of the mergers [N. E. Drakos et al. \(2019a\)](#). as seen in Figure 1. Although the relationship between energy of a system and the spread (how compacted/extended) of a remnant appears to be direct, the correlation between the energy of a system and the radius of the scaled density profile of the remnant appears to be an inverse relationship [N. E. Drakos et al. \(2019b\)](#).

A lot of the understanding of dark matter halo mergers comes from simulations done with equal mass binary systems. This is a simplified setting for mergers, as it does not take into account mergers between different sized halos or mergers between more than two halos. These simulations do not take into account halo substructures [N. E. Drakos et al. \(2019a\)](#). One of the largest issues still being examined from these simulations is what causes the central density of merger remnants to decrease, because the current simulations see an increase in central density that does not match the drop seen in central density of halos as they grow [N. E. Drakos et al. \(2019b\)](#). Other questions left open include how the relationships and general findings of these merger simulations might change when including more complicated and more common types of mergers, such as mergers between more than two halos.

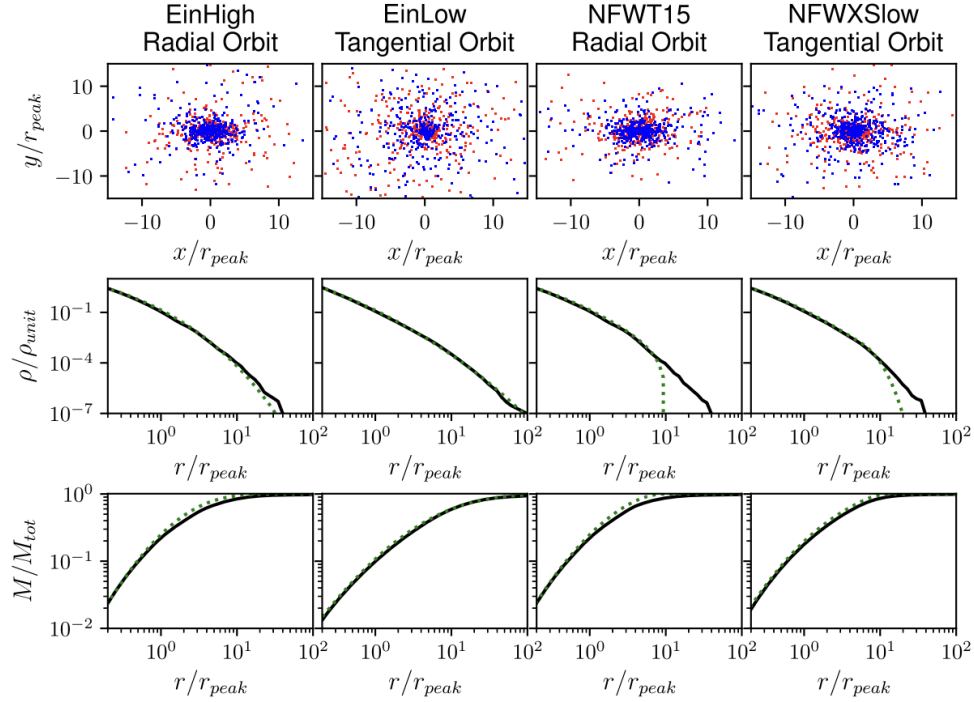


Figure 1. Figure 10 from Drakos + MNRAS, 2019a, 487, 993. This figure shows the results from the paper’s merger simulations. The top graphs are the particles of the remnants, the middle graphs are the density profiles, and the bottom graphs are the mass profiles. These results show the relations between radial orbits and prolate shapes, and tangential orbits and oblate shapes. This figure confirms the idea that the shape of the remnant is dependent on the spin of the mergers.

2. PROJECT

In this paper, we will examine the change in the shape of the remnant during the collision, as well examining how the contributions from the two original galaxies disperse during and after the merger.

This project will address the question of if mergers in complicated systems with more than one galaxy influencing the merger, will coincide with the findings from the simulations of simplified mergers. It will also look into how the particles from a specific galaxy diffuse through the remnant after the merger.

Understanding what factors influence the shape of the remnants is important to galaxy evolution because it helps us understand how the galaxies in our universe evolve over time. Understanding merger remnant shapes can help us to understand the future of younger galaxies, as well as give us insights into older galaxies. Overall, understanding the shape of the merger gives us insight into the life cycles of galaxies.

3. METHODOLOGY

The simulations used in this project are from the work done on interactions between the Milky Way galaxy and the LMC (Large Magellanic Cloud) and SMC (Small Magellanic Cloud). G. Besla et al. (2012) These simulations are done using an N-body simulation which is a simulation of a system of particles and the forces acting on and between the particles in that system. The N-body system used in Marel, Besla +2012 employs smoothed-particle hydrodynamic code, and is modeled with three different types of particles from the stellar bulge, stellar disk, and dark matter halo. This simulation runs approximately 12 Gyr.

The particle types this research will be using for analysis are the Type 1 dark matter or halo particles. This research will be looking at lower resolution (VLowRes) snapshots after the two galaxies merge, so in roughly 6.5 Gyr, which correlates to snapshots 435 and beyond. This research will compare multiple snapshots after merger, including snapshots 435, 535, 635, and 735 to compare how the shape of the remnant as well as the dispersion of the particles from the two original galaxies, changes over time.

One way the questions of this research gets addressed is by examining dark matter distribution 2-dimensional histograms in the XY, YZ, and XZ dimensions at the different snapshots. The three histograms for each snapshot have ellipses fitted to them that give the axes for the shape of the remnant at that time. Creating ratios of these axes determine if the remnant is oblate or prolate. Computing these ratios for each of the snapshots being used provides us the shape of the remnant over time.

The histograms used for the triaxiality mentioned in the paragraph above is one of the sets of plots being created. The ratio of the axes from these histograms is also plotted over time to show how the shape of the remnant changes over time. An additional plot shows the particle contributions from each of the original galaxies over time. The way this is done is by making another histogram of the remnant that shows the ratio of Milky Way particles to M31 particles. The histogram of this ratio illustrates the areas dominated by Milky Way particles and the areas dominated by M31 particles in the remnant. Generating this histogram for the different snapshots shows us how the contributions from each galaxy get distributed over time after the merger. The models below represent the questions being investigated by this project:

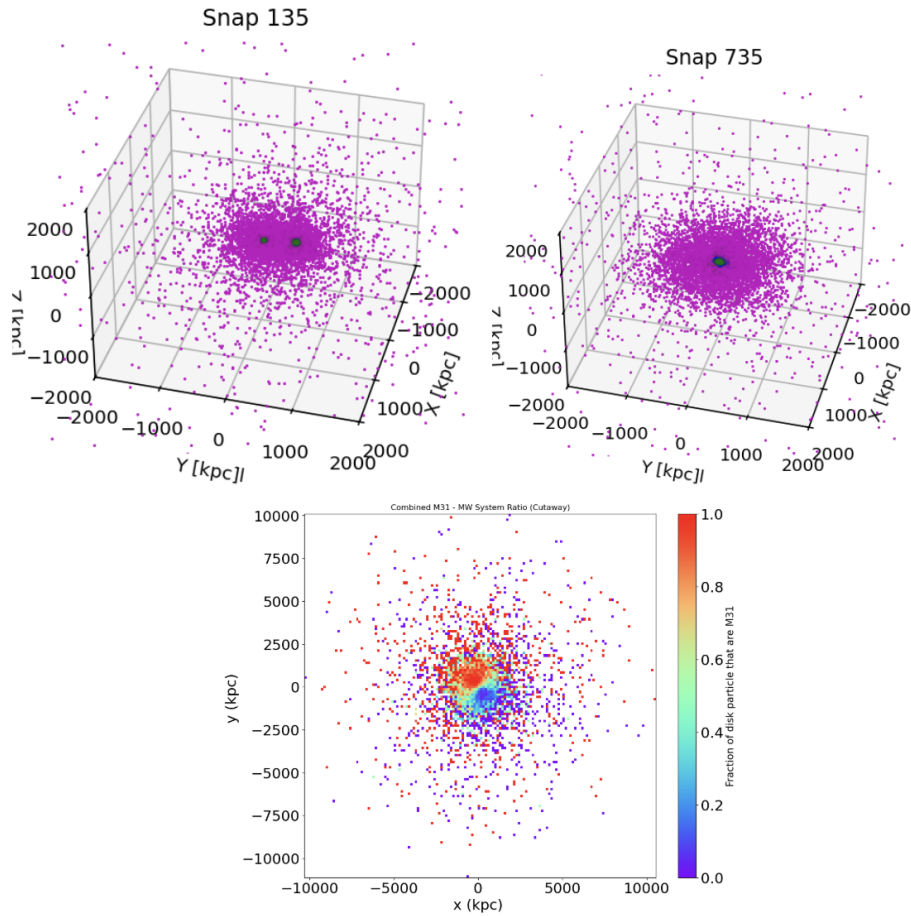


Figure 2. The two figures at the top are three dimensional plots of a sample of the particles before the merger(left) and after the merger(right). These illustrate the investigation this project is doing into the shape of the remnant. The figure at the bottom is a histogram of the ratio of M31 to Milky Way particles. This illustrates the project's examination of the contributions from the two original galaxies to the final remnant.

Due to the path of collision the Milky Way and M31 are on, the expected outcome of this research is to find a 3-dimensional simulation with the qualities similar to the radial orbit remnants shown in Figure 10 from Drakos et al. The anticipated end result will have particles from each of the galaxies intermixing, but with one galaxy's particle type dominating the center of the remnant while the other galaxy's particle types become more prominent the further you get from the center of the remnant.

4. RESULTS

Figure 3 shows the fitted ellipses of the remnant. The ellipse of the remnant in the XY is on the left, the XZ is in the middle, and the YZ is on the right. The difference between the axes will determine if the shape of the remnant is prolate or oblate. If one axis dominates over the other two, the object is considered prolate, if two axes are dominating over the other one, the object is oblate. *N. E. Drakos et al. (2019a)* From Figure 3 we can see that the XY and XZ axes dominate over the YZ. According to the definition of triaxiality, this domination makes the shape of the remnant oblate.

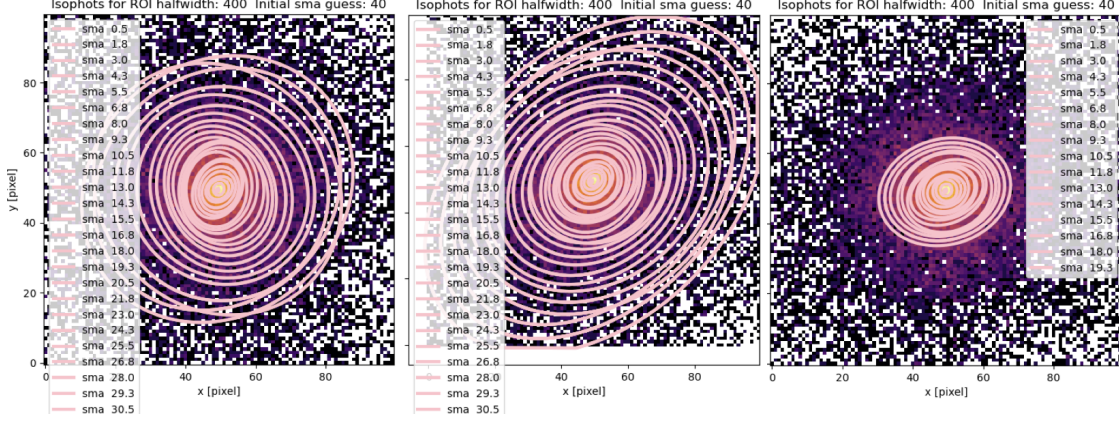


Figure 3. The figure shows the fitted ellipses of the XY(left), XZ(center), YZ(right). The ellipses are generated with bins of 400 and an initial semi-major axis guess of 40. Counting the ellipses fits and comparing the three dimensions we can find that the XY and XZ dominate over the YZ making the remnant oblate.

Figure 4 shows the histogram of the ratio of Milky Way to M31 particles. This illustrates what areas of the remnant are dominated by the Milky Way particles and what areas are dominated by the M31 particles. Figure 4 shows these ratio histograms throughout different snapshots. This shows how the contributions from the two original galaxies changes during the merger. The figure shows histograms from snapshots: 135, 235, 335, 435, 535, 635, 735. Figure 4 displays that although the particles from the Milky Way and M31 combine during the remnant, overall the contributions from each original galaxy remain distinct and do not intermix completely.

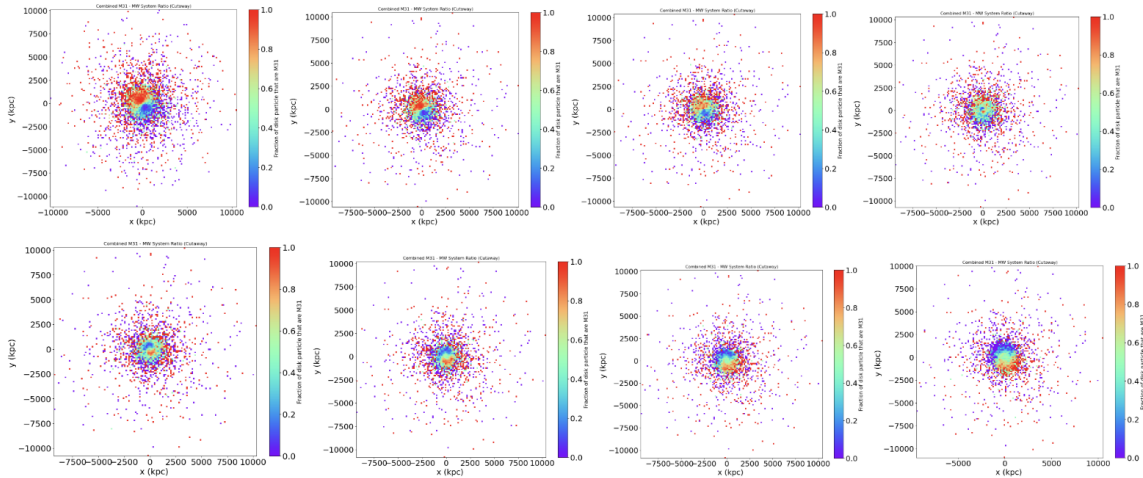


Figure 4. The figure shows the particle distribution over time. The color gradient goes from the red end which is M31 dominated, to the blue end which Milky Way dominated. This figure shows that full intermixing does not occur and comparable sections of the remnant remain dominated by each of the original galaxies' particles.

5. DISCUSSION

Figure 3 Shows that the remnant of the merger between the Milky Way and the M31 merger will have a shape that has a triaxiality that is considered oblate. This goes against the initial hypothesis that it will act like radial orbits from Drakos et al Figure 10, which was prolate.

Results from Figure 3 relates to the Drakos et al paper because it is oblate which is in line with what the paper claims should be the result of a tangential orbit. This finding helps to further understanding of galaxy evolution because it helps determine the generalizations that can be made about the shapes of remnants after mergers.

The uncertainty in Figure 3 analysis comes mainly from the fitting of the ellipses in each plane. The photutils ellipse fitting did not have to fit the ellipse to match the fit of the plane. This is exemplified in Figure 4 by the ellipses that do not match the scale of the plane directly. These fits could lead to errors in determining the lengths of the semi-major axes. This uncertainty leads to some uncertainty about the triaxiality found of the final remnant. . Figure 4 shows that although there is some intermixing of particles from the two original galaxies during the merger, they tend to stay grouped mainly in sections corresponding to their original galaxy. The two sections with particles dominating from Milky Way and M31 stayed on separate sides of the remnant, and the center of the remnant was where the most equal intermixing occurred. This disproved the hypothesis that one galaxy's particles would dominate in the center.

The results from Figure 4 correspond with the paper by Besla et al because the N-body simulation used to simulated this merger means the particles from the two original galaxies would still have angular momentum and velocity vectors related to other particles from the original galaxy. This means most particles from each galaxy would stay grouped together.

The uncertainties from Figure 4 analysis come from the fact that the analysis is almost completely qualitative. Although the computing the ratio is quantitative, the overall results taken from the histograms is from qualitatively looking at the areas of domination. Further quantitative analysis looking in the size of the Milky Way and M31 dominated portions of the remnant and how they compare numerically could be done to further this analysis.

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